

RO-CAL 0.25mcg

PRODUCT DESCRIPTION

Each capsule contains calcitriol 0.25mcg.

Appearance : Two pieces, white, size '2' hard vegetable capsules containing yellowish semisolid mass.

PHARMACODYNAMIC

The two known sites of action of calcitriol are intestine and bone. Evidence also suggests that calcitriol may also act on the kidney and the parathyroid glands. Calcitriol is an active form of vitamin D3. It stimulates intestinal calcium transport thus increasing calcium absorption. The kidneys of uremic patients cannot adequately synthesize calcitriol, the active hormone formed from precursor vitamin D. Hypocalcemia and secondary hyperparathyroidism are a major cause of the metabolic bone disease of renal failure due to inability to synthesize calcitriol. In renal osteodystrophy calcitriol appears to correct hypocalcemia and secondary hyperparathyroidism. Calcitriol treatment is not associated with an accelerated rate of renal function deterioration. Radiographic evidence of extraskelatal calcification has not been found in predialysis patients. Pharmacologic activity duration of a single dose of calcitriol is about 3 to 5 days.

PHARMACOKINETICS

Absorption

Calcitriol is rapidly absorbed from the intestine. Peak serum concentrations is between 3 to 6 hours following oral administration of single doses of 0.25 to 1.0 mcg.

Following multiple-dose administration, steady-state is reached within 7 days.

Distribution

Calcitriol is approximately 99.9% bound in blood. Calcitriol and other vitamin D metabolites are transported in blood, by an alpha-globulin vitamin D binding protein. Maternal calcitriol may enter the fetal circulation. Calcitriol is found into human breast milk at low levels.

Metabolism

Metabolism takes place in the kidneys and intestines, resulting in metabolites such as 1 α , 25R(OH)2-26, 23S-lactone D3 1 α , 25(OH)2-24-oxo-D3; 1 α , 23,25(OH)3-24-oxo-D3; 1 α , 24R,25(OH)3D3,1 α , 25S,26(OH)3D3, 1 α , 25(OH)2-23-oxo-D3; 1 α , 25R,26(OH)3-23-oxo-D3; 1 α , (OH)24,25,26,27-tetranor-COOH-D3.

Excretion

Calcitriol goes through enterohepatic circulation and is excreted in the bile. The metabolites of calcitriol are excreted primarily in faeces and to a lesser amount in the urine. The elimination half-life of calcitriol in serum after single oral doses is about 5 to 8 hours in normal subjects.

INDICATION

Established postmenopausal osteoporosis.

Renal osteodystrophy in patients with chronic renal failure, particularly those undergoing hemodialysis.

Management of hypocalcemia and its clinical manifestations in patients with postsurgical hypoparathyroidism, idiopathic hypoparathyroidism, and pseudohypoparathyroidism. Vitamin D dependent rickets.

Hypophosphatemic vitamin – D resistant rickets.

Predialysis Patients : Management of secondary hyperparathyroidism and resultant metabolic bone disease in patients with moderate to severe chronic renal failure (Cr 15 to 55 mL/min) not yet on dialysis.

Dialysis Patients : Management of hypocalcemia and the resultant metabolic bone disease in patients undergoing chronic renal dialysis.

DOSAGE AND ADMINISTRATION

The optimal daily dose must be carefully determined. Therapy should always be started at the lowest possible dose and should not be increased without careful monitoring of serum calcium.

The effectiveness of therapy is predicated on the assumption that each patient is receiving an adequate but not excessive daily intake of calcium. Patients are advised to have a dietary intake of calcium at a minimum of 600mg daily and not exceed 1000mg.

During the titration period, serum calcium levels should be checked at least twice a week. When the optimal dose has been determined, serum calcium levels should be checked every month (or as given below for individual indications). Samples for serum calcium estimation should be taken without a tourniquet.

Should the serum calcium levels rise to 1 mg/ 100ml (250 μ mol/l) above normal (9 to 11 mg/100 ml or 2250 - 2750 μ mol/l), or serum creatinine rises to > 120 μ mol/l, treatment should be stopped immediately until normocalcaemia ensues.

Most patients respond to between 0.5 mcg and 1.0 mcg daily. (Also refer to drug interaction section if there are any concomitant drugs).

An oral pulse therapy with an initial dosage of 0.1 mcg/kg/week split into two or three equal doses given at the end of the dialysis has been shown to be effective. A maximum total cumulative dosage of 12 mcg per week should not be exceeded.

Post-menopausal Osteoporosis

The recommended dose is 0.25 mcg twice daily. Serum calcium and creatinine levels should be determined at 1, 3 and 6 months and at 6 monthly intervals thereafter.

Dialysis Patients

The recommended initial dose is 0.25 mcg/day. If no satisfactory response in the biochemical parameters and clinical manifestations of the disease state is observed, dosage may be increased by 0.25 mcg/day at 4 to 8 week intervals. During this titration period, serum calcium levels should be obtained at least twice weekly, and if hypercalcemia is noted, the drug should be immediately discontinued until normocalcemia ensues. Phosphorus, magnesium and alkaline phosphatase should be determined periodically.

Patients with normal or only slightly reduced serum calcium levels may respond to doses of 0.25 mcg every other day. Most patients undergoing hemodialysis respond to doses between 0.5 and 1 mcg/day.

Hypoparathyroidism

The recommended initial dosage of 0.25 mcg/day given in the morning. If a satisfactory response in the biochemical parameters and clinical manifestations of the disease is not observed, the dose may be increased at 2- to 4-week intervals. During the dosage titration period, serum calcium levels should be obtained at least twice weekly and if hypercalcemia is noted, calcitriol should be immediately discontinued until normocalcemia ensues. Careful consideration should also be given to lowering the dietary calcium intake. Serum calcium, phosphorus, and 24-hour urinary calcium should be determined periodically. Malabsorption is occasionally noted in patients with hypoparathyroidism; hence, larger doses may be needed.

MODE OF ADMINISTRATION

Oral

CONTRAINDICATIONS

Should not be given to patients with hypercalcemia or evidence of vitamin D toxicity and in patients with known hypersensitivity to calcitriol (or drugs of the same class) or any of the excipients.

WARNINGS AND PRECAUTIONS

If the patient is switched from a long acting vitamin D preparation (e.g. ergocalciferol or colecalciferol) to calcitriol, it may take several months for the level in the blood to return to the baseline value, thereby increasing the risk of hypercalcaemia.

Intake of all other vitamin D compounds and their derivatives, including proprietary compounds or foodstuffs which may be "fortified" with vitamin D, should be monitored. An abrupt increase in calcium intake as a result of changes in diet (e.g. Increased consumption of dairy products) or uncontrolled intake of calcium preparations may trigger hypercalcemia. Patients and families should be advised that strict adherence to prescribed diets is mandatory. Patients and their caregivers should also be carefully informed about the symptoms of hypercalcemia.

In patients with end-stage renal failure, there is a need to monitor the plasma phosphate level.

Immobilised patients, e.g. those who have undergone surgery, are particularly exposed to the risk of hypercalcemia.

Patients with normal renal function who are taking calcitriol should avoid dehydration. Adequate fluid intake should be maintained.

Should hypercalcemia develop, treatment should be stopped immediately. During periods of hypercalcemia, serum calcium and phosphate levels must be determined daily. When normal levels have been attained, treatment can be continued, at a daily dose 0.25 mcg lower than that previously used.

DRUG INTERACTION

Vitamin D and its derivatives should be withheld during treatment to avoid possible additive effects and hypercalcemia.

Calcium supplement intake should be observed strictly and uncontrolled intake of additional calcium containing preparations should be avoided.

Thiazide diuretic increases the risk of hypercalcaemia.

Calcitriol dosage must be determined with care inpatients taking digitalis, as hypercalcemia in such patients may precipitate cardiac arrhythmias.

Corticosteroids inhibit calcium absorption.

Vitamin D derivatives can increase magnesium absorption, although magnesium balance is generally not affected, owing to a compensatory increase in urinary excretion. Magnesium containing drugs (e.g. antacids) are not recommended to be taken by patients with chronic renal failure on dialysis since hypermagnesaemia could occur.

Phosphate-Binding Agents : Since calcitriol has an effect on phosphate transport in the intestine, kidneys and bones, the dosage of phosphate-binding agents must be adjusted in accordance with the serum phosphate concentration.

Enzyme inducers such as phenytoin or phenobarbital may lead to increased metabolism leading to reduced serum concentrations of calcitriol. Higher doses of calcitriol may be required.

Colestyramine can reduce intestinal absorption of fat-soluble vitamins and therefore may impair intestinal absorption of calcitriol.

Ketoconazole may inhibit both synthetic and catabolic enzymes of calcitriol.

PREGNANCY AND LACTATION

The safety of calcitriol during pregnancy has not been established. Studies of reproductive toxicology in animals have not yielded unequivocal findings, and no controlled studies on the effect of exogenous calcitriol on pregnancy and foetal development have been performed in human subjects. Consequently calcitriol should be given only when the potential benefit has been weighed against the possible hazard to the foetus. The usual caution in prescribing any drug for women of childbearing age should be observed.

It should be assumed that exogenous calcitriol passes into breast milk. Mothers may breastfeed while taking calcitriol, provided that the serum calcium levels of the mother and infant are monitored.

SIDE EFFECTS

Adverse effects are, in general, similar to those encountered with excessive vitamin D intake, ie, hypercalcemia syndrome or calcium intoxication. Due to the short biological half-life of calcitriol, pharmacokinetic investigations have shown normalization of elevated serum calcium within a few days of treatment withdrawal.

Signs and symptoms of vitamin D intoxication associated with hypercalcemia include: *Early*: weakness, headache, somnolence, nausea, vomiting, dry mouth, constipation, muscle pain, bone pain, metallic taste, and anorexia, abdominal pain or stomach ache. *Late*: polyuria, polydipsia, anorexia, weight loss, nocturia, conjunctivitis (calcific), pancreatitis, photophobia, rhinorrhea, pruritus, hyperthermia, decreased libido, elevated BUN, albuminuria, hypercholesterolemia, elevated SGOT (AST) and SGPT (ALT), ectopic calcification, nephrocalcinosis, hypertension, cardiac arrhythmias, dystrophy, sensory disturbances, dehydration, apathy, arrested growth, urinary tract infections, and, rarely, overt psychosis. Hypersensitivity reactions (pruritus, rash, urticaria, and very rarely severe erythematous skin disorders) may occur in susceptible individuals.

SYMPTOMS AND TREATMENT OF OVERDOSAGE

Symptoms : anorexia, headache, vomiting, constipation, dystrophy, sensory disturbance, fever, thirst, polyuria, dehydration, apathy, urinary tract infection, hypercalcemia.

The treatment of overdosage should consist of general supportive measures. If drug ingestion is discovered within a relatively short time induction of emesis or gastric lavage may be of benefit in preventing further absorption. If the drug has passed through the stomach, the administration of mineral oil may promote its fecal elimination. Calcium serum electrolyte determinations, rate of urinary calcium excretion, and assessment of electrocardiographic abnormalities due to hypercalcemia should be obtained. Should persistent and markedly elevated serum calcium levels occur, therapeutic alternatives may be considered, such as the use of drugs such as phosphates and corticosteroids as well as measures to induce an appropriate forced diuresis.

STORAGE CONDITIONS

Store below 30°C. Store in the original package and keep the blister in the outer carton in order to protect from light and moisture

SHELF LIFE

3 years

PACK SIZE

Blister of 10 capsules. Box of 1, 2, 3, 6, 10, 15, 20, 25 and 50 blisters.

MANUFACTURED BY / FOR:

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